

Project Concept Note: Vibe Coding Final Project

Project Title / Application Name

CareerPilot AI

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Live Application URL: <https://careerpilot-ai-plum.vercel.app/>

Problem Statement / Objective

In today's competitive job market, professionals and job seekers often struggle to align their existing skills with target roles, optimize their resumes for Applicant Tracking Systems (ATS), and prepare effectively for behavioral or technical interviews. The objective of CareerPilot AI is to solve this by providing a unified, AI-powered toolkit that guides users through every stage of the career advancement process—from initial career discovery to final interview preparation.

Target User and Use Case

- **Target Users:** Job seekers, recent graduates, and professionals looking to transition into new careers or elevate their current trajectory.
- **Use Case:** A user uploads their resume and pastes a job description for a role they want. The application instantly analyzes the gap between their current resume and the job requirements, suggests targeted improvements, and then acts as an AI coach to conduct a mock interview based on that specific role.

LLM Model and API Used

- **Model:** Google Gemini API
- **SDK:** `google-generativeai` (Python)
- The backend relies on Gemini's fast inference and streaming capabilities to deliver real-time feedback and analysis to the frontend.

Key Features of the Application

1. **Career Discovery:** Suggests alternative career paths and identifies skill gaps based on the user's background and interests.
2. **Resume Analyzer:** Accepts PDF uploads, extracts the text, scores the resume, identifies red flags, and suggests ATS-friendly rewrites.
3. **JD (Job Description) Analyzer:** Extracts core responsibilities, required hard/soft skills, and hidden requirements from dense job descriptions.
4. **Interview Coach:** Generates hyper-specific mock interview questions (HR, Technical, Behavioral) and provides model answers based on the target role context.

Expected User Experience and Outcomes

- **User Experience:** The user will experience a fast, modern, and responsive interface (built with React and Tailwind CSS). Because the AI responses are streamed directly from the backend, users will see the AI's analysis type out in real-time, eliminating frustrating loading screens.
- **Outcomes:** Users will walk away with a highly optimized resume tailored to their target job, a clear understanding of what skills they need to learn, and the confidence to ace their upcoming interviews through targeted AI coaching.